

Example 4: Let  $T$  be the linear transformation whose standard matrix is

$$A = \begin{bmatrix} 1 & -4 & 8 & 1 \\ 0 & 2 & -1 & 3 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

Does  $T$  map  $\mathbb{R}^4$  onto  $\mathbb{R}^3$ ? Is  $T$  a one-to-one mapping?

Solution: By defn. of onto and image of  $x$ , For every vector  $b$  in  $\mathbb{R}^3$ , the equation  $T(x) = b$  has at least one solution, where  $x \in \mathbb{R}^4$ .

Since this is a matrix transformation,  $\therefore$  We need to check whether the matrix equation  $Ax = b$  has at least one solution for every  $b$ .

$$\begin{aligned} & \begin{bmatrix} 1 & -4 & 8 & 1 \\ 0 & 2 & -1 & 3 \\ 0 & 0 & 0 & 5 \end{bmatrix} \xrightarrow[\times 1/5]{\text{Row 3} = \text{Row 3}} \begin{bmatrix} 1 & -4 & 8 & 1 \\ 0 & 2 & -1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow[\text{Row 1} = \text{Row 1}]{-\text{Row 3}} \begin{bmatrix} 1 & -4 & 8 & 0 \\ 0 & 2 & -1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ & \xrightarrow[\times \text{Row 3}]{\text{Row 2} = \text{Row 2} - 3} \begin{bmatrix} 1 & -4 & 8 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow[\text{Row 1} = \text{Row 1}]{2 \times \text{Row 2}} \begin{bmatrix} 1 & 0 & 6 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow[\times 1/2]{\text{Row 2} = \text{Row 2}} \begin{bmatrix} 1 & 0 & 6 & 0 \\ 0 & 1 & -1/2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

$$\begin{bmatrix} 1 & 0 & 6 & 0 \\ 0 & 1 & -1/2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Verification of whether the final matrix is in ref

1. Row 1 to 3 are all non-zero rows.  $\therefore$  The condition that all non-zero rows must be above the zero rows is vacuously True.

2. Row 1 Leading entry = 1 in Column 1  
Row 2 Leading entry = 1 in Column 2  
Row 3 Leading entry = 1 in Column 4

$\therefore$  The condition that for each row, the leading element must be in a column to the right of the columns of the leading entries of all rows above it is True.

3. Row 2 Column 1 = Row 3 Column 1 = 0  
Row 3 Column 2 = 0

$\therefore$  The condition that for each column having a leading entry, all positions below it must contain 0 is True.